

AI Exposure in a Dual Labor Market: Evidence from Côte d’Ivoire’s National Employment Survey

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Draft — August 2026

Abstract. How exposed are workers in economies with predominantly informal employment to the labor market disruptions of artificial intelligence? This paper maps AI occupational exposure onto Côte d’Ivoire’s labor force using 4-digit-ISCO-08-coded micro-data from the Enquête Nationale sur l’Emploi auprès des Ménages (ENE-M), applying the GPT exposure scores of Eloundou et al. (2024) and the AI Occupational Exposure index of Felten, Raj, and Seamans (2021); a third index (Pizzinelli et al. 2023) is discussed in the literature but not yet integrated empirically. I propose a digitally-reachable (readiness-adjusted) exposure measure that scales occupational exposure by predicted workplace digital-infrastructure access, using ENE-M and DATAFID (ANStat’s ICT-access survey); because this adjustment is built substantially from the same occupation, formality, and education variables as exposure itself, it is presented explicitly as an infrastructure-readiness index rather than a re-measurement of task content — and, given known asymmetries in what its two specifications capture (Section 4.4), not as a precise bound either — with a separate task-content verification (using ENE-M’s free-text job description field) proposed as the more direct test of that claim. This paper’s central empirical results — the magnitude of the readiness adjustment and its distribution across occupations and regions — are not yet computed as of this draft; the contribution reported here is the measurement framework, its data sources, and a first set of usage-side findings. Country-level usage data from two independent AI providers — the Anthropic Economic Index (Claude) and OpenAI Signals (ChatGPT) — show that Côte d’Ivoire ranks in the lowest usage quartile on both, but that the two providers report usage on different, not directly comparable descriptors (Claude’s classified usage by occupation, skewing toward IT; ChatGPT’s usage by topic, skewing toward general writing and everyday tasks), which cautions against treating a single provider’s usage data as a general statement about “AI usage” in a country; linking this usage data to the ENE-M’s occupational distribution directly (via a planned Kitagawa-Blinder-Oaxaca decomposition) is proposed but not yet executed. The paper argues that AI exposure indices, developed for advanced economies, require adaptation — and a clearer separation of task-content exposure from infrastructure readiness — before they can guide labor market policy in countries with dual employment structures.

Keywords: artificial intelligence, labor market exposure, informal employment, Côte d’Ivoire, occupational classification, Anthropic Economic Index, OpenAI Signals

JEL codes: J46, O33, O15, O55, E26

1. Introduction

The rapid advance of large language models (LLMs) and other AI systems has prompted a surge of research on their potential impact on labor markets. A now-standard empirical approach consists of measuring the “exposure” of occupations to AI capabilities — that is, the share of tasks within an occupation that AI systems could, in principle, perform or substantially accelerate. The seminal contributions of Felten, Raj, and Seamans (2021), Eloundou et al. (2024), and Pizzinelli et al. (2023) have produced occupation-level exposure scores that are widely used in policy analysis by the IMF, World Bank, ILO, and OECD, among others.

However, these indices share a common empirical foundation: the O*NET database maintained by the U.S. Bureau of Labor Statistics. O*NET describes the task content of occupations as they are performed in the United States — a high-income, predominantly formal economy. When these indices are applied to other countries via crosswalks to the International Standard Classification of Occupations (ISCO-08), an implicit assumption is made: that a given ISCO code maps to roughly the same bundle of tasks regardless of the country or institutional context in which the occupation is performed. This assumption is unlikely to hold in economies where the dominant form of employment is informal.

This paper examines what happens when AI exposure indices are applied to a labor market where informal employment accounts for roughly 90 percent of total employment. Côte d’Ivoire provides an instructive case. Its Enquête Nationale sur l’Emploi auprès des Ménages (ENE-M) is a nationally representative panel survey coded at the 4-digit ISCO-08 level, with detailed information on employment formality, working conditions, and — critically — access to information and communication technologies (ICTs) in the workplace. These data make it possible not only to map standard AI exposure indices onto the Ivorian occupational distribution, but also to construct a digital-reachability adjustment (Section 4.4) that scales exposure by workplace digital-infrastructure access — for the GPT-based scores, this mechanically bounds the constructed measure from above (Section 5.2.3), though not necessarily the true reachable exposure it approximates (Section 6.1) — a step toward, though not on its own a full test of, the paper’s broader claim that the theoretical and actual task content of occupations diverge in the informal sector.

The paper makes three contributions. First, it provides the first AI exposure mapping for a francophone West African economy, extending the geographic coverage of the existing literature beyond the six countries analyzed by Pizzinelli et al. (2023), the 25 countries covered by Demombynes, Langbein, and Weber (2025), the region-wide analysis of IMF (2026), and the country-level analyses produced by the Anthropic Economic Index team. Unlike these continent- or region-wide studies, which necessarily rely on harmonized cross-country data of limited occupational granularity, this paper exploits the full 4-digit ISCO-08 detail and the formality, sector, and ICT-access variables of a single country’s labor force survey — a depth that region-wide comparative work cannot offer for any individual country, complementing rather than substituting for it. Second, it proposes a simple and replicable method for adjusting AI exposure indices for digital-infrastructure readiness, using ICT access as a proxy for whether an occupation’s O*NET-described task content is currently reachable in a given work environment. This contribution is methodological: the adjustment can be applied to any country with a labor force survey that captures both ISCO codes and formality status; as discussed in Section 4.4, its GPT-based component mechanically bounds the *constructed measure* from above, but neither of its two specifications should be read as a reliable bound on *true* reachable exposure, and neither substitutes for a direct re-measurement of task content. Third, it juxtaposes theoretical AI exposure with observed AI usage in Côte d’Ivoire using publicly available

usage data from two independent AI providers — the Anthropic Economic Index and OpenAI Signals — providing country-level evidence on the scale of AI diffusion in a low-income francophone context, and illustrating, through the difference between the two providers’ usage composition, that single-provider usage data can reflect the provider’s user base as much as the country’s overall AI usage pattern; directly linking this usage evidence to the ENE-M’s occupational micro-data via the planned decomposition in Section 4.5 is the natural next step, not yet completed in this draft.

The paper proceeds as follows. Section 2 provides background on the Ivorian labor market and the ENE-M survey. Section 3 reviews the AI exposure literature and its application to developing countries. Section 4 describes the data and methodology. Section 5 presents results and, in Section 5.4, consolidates robustness checks and open measurement caveats. Section 6 discusses implications, and Section 7 concludes.

2. Background: the Ivorian labor market

2.1 Structural features

Côte d’Ivoire is the largest economy in the West African Economic and Monetary Union (WAEMU), with a GDP per capita of approximately USD 2,600 (2023). Despite sustained economic growth averaging over 7 percent annually since 2012, the labor market remains characterized by several structural features that distinguish it sharply from the advanced economies on which AI exposure indices are calibrated.

The most salient feature is the predominance of informal employment. According to the ENE 2019 and subsequent rounds, approximately 89 to 93 percent of total employment is informal, depending on the definition used (enterprise-based vs. job-based, following the ILO 2023 guidelines from the 21st ICLS). The informal employment rate varies substantially by milieu de résidence — it is lower (though still dominant) in Abidjan’s formal tertiary sector and approaches near-universality in rural areas. The ENESEI 2016 reported a composite labor underutilization indicator (SU4) of 27.8 percent, underscoring that the low headline unemployment rate (around 3 percent under the strict ILO SU1 definition) masks significant underemployment and labor market slack.

The occupational structure, coded under ISCO-08, is concentrated in three major groups: Group 5 (service and sales workers), Group 6 (skilled agricultural workers), and Group 9 (elementary occupations). Groups 1 through 3 — managers, professionals, and technicians — which are the most AI-exposed categories in high-income countries, account for a comparatively small share of total employment. This compositional difference alone would predict lower aggregate AI exposure in Côte d’Ivoire relative to advanced economies, a finding consistent with the cross-country pattern documented by Cazzaniga et al. (2024) and Pizzinelli et al. (2023).

A second distinguishing feature is the limited penetration of digital technologies in the workplace. While mobile phone ownership is widespread, access to computers and the internet in a work context remains confined to a narrow segment of the formal sector. DATAFID (Section 4.1) and the EHCVM (Enquête Harmonisée sur les Conditions de Vie des Ménages) provide complementary data on this dimension. Section 3.2 develops the conceptual implication of this digital access gap for AI exposure measurement — that it can alter the effective task content of an occupation, not merely constrain AI adoption directly.

2.2 The ENE-M survey

The Enquête Nationale sur l’Emploi auprès des Ménages (ENE-M) is ANStat’s flagship labor force survey, designed as a rotating panel with a 2-2-2 rotation scheme across 1,896 primary sampling units (zones de dénombrement) spanning all 33 administrative regions. The survey instrument is aligned with the CIST (Conférence Internationale des Statisticiens du Travail) resolutions and the ILO guidelines for labor force surveys, and produces the full suite of labor underutilization indicators from SU1 (strict unemployment) through SU4 (composite underutilization including time-related underemployment and the potential labor force).

For the purposes of this paper, the key variables are:

- **Occupation:** coded at the 4-digit ISCO-08 level, enabling direct linkage with AI exposure scores via the ISCO-SOC crosswalk
- **Employment formality status:** following the ILO/CIST criteria (enterprise registration, written contract, social security contributions), allowing decomposition of each occupational category into formal and informal segments
- **Sector of activity:** coded under ISIC Rev.4 (CITI Rev.4)
- **ICT access variables:** access to computer at work, internet use for work purposes, mobile phone use for professional activities
- **Standard demographic and human capital variables:** age, sex, education level, region, urban/rural

The rotating panel design allows, in principle, the observation of labor market transitions, although this paper focuses on the cross-sectional dimension for the exposure mapping exercise. The panel structure could be exploited in future work to study transitions between more and less AI-exposed occupations over time.

3. AI exposure indices: review and applicability to developing countries

3.1 The standard approach

The measurement of AI occupational exposure has converged around a common methodology: linking AI capabilities to the task content of occupations as described in the O*NET database, then aggregating task-level exposure scores to the occupation level.

Felten, Raj, and Seamans (2021) introduced the AI Occupational Exposure (AIOE) index, which maps advances in specific AI application areas (image recognition, language modeling, etc.) to the abilities required by different occupations, as listed in O*NET. The AIOE captures the breadth of AI capabilities relevant to each occupation.

Eloundou et al. (2024) developed the GPT exposure framework, which asks: for each Detailed Work Activity (DWA) in O*NET, can a GPT-4-class LLM reduce the time required to complete the task by at least 50 percent, either directly (E1 exposure) or with access to complementary software tools (E2 exposure)? Using both human annotators and GPT-4 as classifiers, they produced occupation-level exposure scores ranging from 0 to 1. A key finding is that approximately 80 percent of U.S. workers have at least 10 percent of their tasks exposed, and about 19 percent have at least 50 percent of their tasks exposed.

Pizzinelli et al. (2023) extended the Felten et al. framework to six countries (US, UK, Brazil,

Colombia, India, South Africa) using the ISCO-SOC crosswalk and worker-level microdata. Their innovation is the distinction between AI as a complement (augmenting worker productivity) and AI as a substitute (displacing worker labor), introducing a “high complementarity” and “low complementarity” decomposition of exposure. They find that advanced economies face higher unadjusted exposure than emerging markets, but that complementarity patterns — and therefore the implications for displacement risk — differ across country income levels.

Most recently, Gmyrek, Berg, and Kamiński (2025) produced a refined ILO index of generative-AI occupational exposure, distinguishing degrees of automation and augmentation potential at the task level. A methodologically relevant feature of this index for the present paper is that it is built directly on ISCO-08 rather than on the U.S. SOC taxonomy, avoiding the many-to-many SOC-to-ISCO crosswalk that Section 4.2 relies on for the other three indices. This makes it a natural candidate for a future crosswalk-free robustness check on the exposure mapping constructed here, although reproducing it is beyond the scope of the present paper and is left as an extension (Section 6.4).

3.2 Limitations for developing country application

Several limitations of the standard approach become apparent when it is applied beyond the OECD context.

The O*NET task content assumption. O*NET describes occupations as they are performed in the United States. The implicit assumption in cross-country applications via the ISCO-SOC crosswalk is that ISCO code X maps to the same task bundle worldwide. This assumption weakens as the institutional context diverges from the U.S. benchmark. In an economy with predominantly informal employment, the actual task content of many occupations differs markedly from the O*NET description: tasks may be less digitized, less codified, more reliant on tacit knowledge and personal relationships, and less amenable to AI augmentation or substitution. Demombynes, Langbein, and Weber (2025) acknowledge this limitation but do not directly address it in their 25-country analysis.

The infrastructure prerequisite. AI exposure is, at best, a measure of technical feasibility — the extent to which an AI system *could* perform or assist with a task. But technical feasibility requires infrastructure: reliable electricity, internet connectivity, access to computing devices. In low-income countries, Demombynes et al. (2025) find that effective exposure is substantially lower than occupational exposure once electricity access is accounted for. This paper extends their approach by using workplace-level ICT access from the ENE-M as a more granular adjustment factor.

The informality gap. The most substantive limitation for countries like Côte d’Ivoire is the treatment of informal employment. Standard exposure indices assign the same score to all workers in a given occupation, regardless of whether they work in the formal or informal sector. The task content of an ISCO 5221 (shop sales assistant) in a formal retail chain — where tasks include operating a computerized point-of-sale system, managing digital inventory, and processing electronic payments — is plausibly different from the task content of an ISCO 5221 in an informal market stall, where transactions are more likely to be in cash and inventory management more likely to be manual. The former is plausibly more exposed to AI-driven automation of checkout, inventory optimization, and customer analytics than the latter; whether the latter is exposed at all — including through smartphone-mediated tools that do not require the point-of-sale infrastructure of a formal retailer, per Section 4.4 — is an empirical question this paper’s planned task-content check (Specification C, Section 4.4) is designed to inform, not one this illustrative comparison settles on its own.

The regional evidence base. Two recent reports bracket the sub-Saharan African evidence this paper speaks to. IMF (2026) provides a region-wide assessment of AI’s potential in sub-Saharan Africa, and AfDB (2025) frames the productivity and inclusion stakes of AI diffusion across the continent; both necessarily operate at a level of aggregation — regional or pan-African — that cannot capture the within-country formal/informal divide documented here for Côte d’Ivoire specifically. This paper’s single-country, survey-microdata approach is intended as a complement to that regional evidence base: it quantifies, at the level of individual ISCO-08 occupations and formality strata, the gap between technical exposure and this paper’s digitally-reachable exposure measure — a gap region-wide exposure figures cannot capture at all, and which this paper’s own caveats (Section 4.4, 5.4) mean should be read as suggestive of, not proof of, an overstatement in region-wide figures specifically.

No existing paper has attempted to systematically adjust AI exposure indices for the formal/informal decomposition of occupations.

3.3 Observed usage vs. theoretical exposure

A parallel literature is emerging on the gap between what AI *could* do (exposure) and what AI *is actually used for* (diffusion). The Anthropic Economic Index, which analyzes millions of real-world conversations with Claude to identify the occupations and tasks for which AI is being used, provides the most granular available data on observed AI usage by occupation and country. The Index’s reports have documented substantial variation in AI usage across countries, with advanced economies showing much higher per-capita adoption than developing economies. A recent IMF working paper by Fan and Nguyen (2026) uses the Anthropic Economic Index data to construct an “AI concentration index” (ACI), finding that in developing economies, virtually all AI-generated economic value is concentrated in a small professional enclave (ACI near 1.0), compared to 0.4–0.5 in high-income economies.

A structurally similar effort exists on the OpenAI side. Chatterji et al. (2025) analyze de-identified ChatGPT conversation data to characterize how people use the product, and OpenAI has since operationalized this pipeline as *OpenAI Signals* (OpenAI, 2026), a public, differentially private data release covering per-capita usage rank, topic composition, and work-relatedness for over 140 countries. Because Claude and ChatGPT have systematically different user bases — Claude’s skewing toward developers and technical professionals, ChatGPT’s toward a broader consumer population, reflecting the roughly order-of-magnitude difference in their respective weekly active user counts — a single-provider usage index risks describing the provider’s user base as much as it describes “AI usage” in a given country. Section 5.3.4 documents this concretely for Côte d’Ivoire by comparing both providers’ available usage metrics side by side; because OpenAI Signals does not publish occupation-level breakdowns outside the United States, this comparison is necessarily limited to country-level metrics (per-capita rank, work-relatedness, topic composition) and cannot itself confirm or refute occupation-level concentration on ChatGPT — a limitation stated explicitly in Section 4.5.

This exposure-usage gap plausibly reflects the factors described above — infrastructure constraints and the informality of employment — and, potentially, a language barrier for francophone Africa, since the dominant LLMs are trained primarily on English-language data; this paper’s own test of the language hypothesis (Section 5.3.3) does not find statistically significant evidence for it, so it is listed here as a candidate mechanism from the literature rather than an established one. The Anthropic Economic Index January 2026 report notes that Claude performs different types of tasks

in countries at different stages of economic development, suggesting that the nature, not just the volume, of AI usage differs across country income levels.

4. Data and methodology

4.1 Data sources

This paper combines seven data sources:

ENE-M micro-data. The pooled ENE-M panel spanning seven quarters, 2024Q3 through 2026Q1 (24T3–26T1), roughly 59,196 observations classified as employed (*actifs occupés*, `en_emp==1`) across all waves. The analysis uses individuals aged 15 and above who are classified as employed under the CIST/ILO framework. Key variables are described in Section 2.2; the construction of the panel-consistent main-job ISCO-08 variable (`citp_final`, 86.9 percent coverage among *actifs occupés*) is documented in the pipeline notes accompanying `scripts/00, 04-11d`. **A sample-construction caveat:** because ENE-M is a rotating panel (2-2-2 rotation, Section 2.2), the pooled 59,196 observations are person-*quarters*, not 59,196 distinct individuals — some respondents appear in multiple quarters. The current cross-sectional exposure mapping treats each person-quarter as an independent observation; this is a simplification the paper adopts for its first pass at the exposure mapping (Section 4.3), and standard errors and aggregate estimates reported here do not yet account for within-person correlation across quarters or for the survey’s stratification/PSU design. Correcting for this — clustering at the individual level, applying quarter-specific survey weights, or restricting to one observation per individual — is listed as a robustness item in Section 5.4 rather than resolved in this draft. The 13.1 percent of *actifs occupés* without a coded `citp_final` are also not yet characterized by formality, region, sector, or education (i.e., whether this residual non-coverage is representative or concentrated in specific segments is not yet checked); this is likewise deferred to Section 5.4.

AI exposure scores. Three sets of occupation-level exposure scores are considered, of which two are applied in this draft:

1. The GPT exposure scores from Eloundou et al. (2024), available at the O*NET SOC level from the authors’ GitHub repository (file: `occ_level1.csv`). These provide E1/alpha (direct LLM exposure), E2/beta and gamma (LLM + software tools exposure), and human- and GPT-rated variants.
2. The AIOE index from Felten, Raj, and Seamans (2021), available from their data appendix (file: `AIOE_DataAppendix.xlsx`).
3. The complementarity-adjusted index from Pizzinelli et al. (2023), which distinguishes high-complementarity and low-complementarity (substitution-risk) exposure. **Not integrated in this draft** — Sections 4.2–4.3 report results for the GPT and AIOE indices only. Pizzinelli et al. is discussed as a literature reference point (Sections 3.1, 6.1) but should not be read as an applied third index until this integration is completed; the ISCO-08-native ILO index of Gmyrek et al. (2025, Section 3.1) is a plausible alternative or complement, since it would not require the SOC crosswalk described in Section 4.2.

ISCO-08 to SOC crosswalk. The concordance table between ISCO-08 codes and 6-digit O*NET-SOC codes, sourced from the Institute for Structural Research (IBS) public crosswalk — used as a substitute for the official BLS ISCO-SOC crosswalk file, which returns a persistent HTTP 403 (anti-bot protection) under automated access. Because the mapping is many-to-many (one ISCO

code can correspond to multiple SOC codes and vice versa), the merge procedure requires the aggregation rules described in Section 4.2.

BLS OEWS national employment, by SOC code. Used to weight the many-to-many crosswalk aggregation in Section 4.2. The official BLS OEWS download pages are subject to the same automated-access block as the ISCO-SOC crosswalk; national cross-industry employment by 6-digit SOC is instead sourced from the May 2020 OEWS national estimates as redistributed by the CRAN package `oews2020` (Mann, 2022). **This file carries a second, distinct instance of the SOC-vintage mismatch documented in Appendix A for the crosswalk itself:** it turns out to use post-2018 SOC codes rather than the SOC-2010 vintage this paper initially assumed, producing an asymmetric weight-coverage rate across the two exposure indices (95.8 percent for GPT, 83.6 percent for AIOE) — full detail, verification, and the fix required are in Section 5.4, not repeated here.

DATAFID 2023 (with EHCVM 2021 as a fallback). DATAFID is ANStat’s dedicated ICT-access survey, covering three linked files: a household/household-head module (`CM`, 1,714 obs.), an enterprise module (`Entreprise`, 462 obs., 284 variables — computer/internet/e-commerce/ICT-investment questions, with survey weights by employment, revenue, and enterprise count), and an individual module (`Individu`, 6,643 obs., 332 variables — individual-level internet and computer use). It is the source for Specification B of the digital-reachability adjustment (Section 4.4); it was chosen over the EHCVM because ICT access is DATAFID’s core subject rather than a minor module. **Its sectoral breakdown is coarser than initially assumed:** the enterprise module’s `SECTEUR` variable distinguishes only three broad categories (primary/secondary/tertiary — 56/187/219 enterprises respectively), not ISIC-section-level detail; Specification B (Section 4.4) is corrected accordingly. The EHCVM 2021 (*Enquête Harmonisée sur les Conditions de Vie des Ménages*) is retained as a fallback household-level ICT-access source.

Anthropic Economic Index data. Publicly available on Hugging Face, the Index provides occupation-level AI usage data (proportion of conversations coded to each O*NET occupation, distinguishing augmentation vs. automation) and country-level usage intensity.

OpenAI Signals data. Publicly available at openai.com/signals under a CC BY 4.0 license (OpenAI, 2026), covering consumer ChatGPT usage (Free/Plus/Pro/Go plans; enterprise and API traffic excluded) from July 2024 to June 2026, differentially private and suppressed below 100 effective messages per cell. Used here for country-level per-capita usage rank, topic composition, and work-relatedness for Côte d’Ivoire, as a cross-provider check on the Anthropic Economic Index (Section 4.5). Unlike the AEI, it does not provide occupation-level (O*NET) breakdowns outside the United States.

4.2 Crosswalk construction

The central methodological challenge is the linkage between the ISCO-08 codes used in the ENE-M and the SOC codes used in the O*NET-based exposure scores. The crosswalk proceeds in three steps:

Step 1: Direct concordance. Using the IBS crosswalk table described in Section 4.1 (the substitute for the inaccessible official BLS crosswalk), each 4-digit ISCO-08 code is matched to one or more 6-digit O*NET-SOC codes. Where the mapping is one-to-one, the exposure score is directly assigned.

Step 2: Aggregation for one-to-many matches. Where one ISCO code maps to multiple SOC

codes, the exposure score assigned to the ISCO code is the employment-weighted average of the constituent SOC scores, using U.S. employment weights from the BLS Occupational Employment and Wage Statistics (OEWS), sourced as described in Section 4.1. This follows the convention in Pizzinelli et al. (2023) and Demombynes et al. (2025). Where a matched SOC code carries no OEWS weight (an aggregated or discontinued detailed-SOC code not covered by the national file), it is dropped from the weighted average for that ISCO code rather than assigned an arbitrary weight; if none of the matched SOC codes for a given ISCO code carry a weight, the unweighted mean is used instead, so no ISCO code is dropped for lack of weighting data alone.

Step 3: Handling unmatched codes. Some ISCO codes used in the ENE-M may not have a clean match in the crosswalk — particularly occupations that are common in sub-Saharan Africa but have no direct U.S. equivalent (e.g., certain agricultural occupations, artisanal trades), as well as, distinctly, ISCO codes affected by the SOC-vintage mismatch described below. As of this draft, unmatched codes are simply excluded (13 of 436, see Appendix A) rather than manually reassigned; task-similarity-based manual assignment is a planned refinement, not yet implemented. **This exclusion is not bias-neutral:** both the agricultural/artisanal codes and the technology-renumbered codes affected by the SOC-vintage issue are plausibly systematically different from the matched sample (in exposure level, formality, or both) rather than a random subset, so dropping them should be expected to shift the aggregate exposure estimate in an as-yet-uncharacterized direction, not treated as a conservative default. A sensitivity analysis quantifying these codes’ ENE-M employment share and its effect on the aggregate is planned (Appendix D, Section 5.4) but not yet run.

A methodologically important caveat to Steps 1–2, not a separate step: the GPT (Eloundou et al.) and AIOE (Felten et al.) scores merged in Step 2 are built on different O*NET-SOC vintages (post-2018 and 2010 respectively), while the crosswalk itself is SOC-2010-vintage; the numeric-code merge used here does not detect this and silently under-matches GPT scores for occupations renumbered in the 2018 SOC revision, concentrated in technology-adjacent occupations. This is documented with the exact affected codes and match-rate impact in Appendix A, and should be read as the crosswalk’s most significant open data-quality issue, not a minor footnote.

The quality of the crosswalk is assessed by reporting: (i) the share of ENE-M employment that is matched vs. unmatched, (ii) the coefficient of variation of exposure scores within each ISCO code (reflecting heterogeneity in the underlying SOC codes), and (iii) the results of the sensitivity analysis. Full crosswalk diagnostics — match rates, OEWS weight coverage, the SOC-vintage issue, and the count of ISCO codes without any exposure score — are reported in Appendix A; the effect of employment weighting relative to a simple unweighted average is reported in Appendix D; a consolidated summary of all measurement caveats is in Section 5.4.

4.3 Standard exposure mapping

Once the crosswalk is constructed, each employed individual i in the ENE-M is assigned an exposure score E_i based on their 4-digit ISCO code. Aggregate exposure measures are then computed as follows.

National aggregate exposure:

$$\bar{E} = \sum_{i=1}^N w_i \cdot E_i$$

where w_i is the survey sampling weight, normalized to sum to 1.

Exposure by subgroup. The same weighted average is computed for subgroups defined by: region (Abidjan, other urban, rural), sex, age group, education level, sector of activity (ISIC section), and — crucially — formality status (formal vs. informal).

Exposure distribution (GPT-based scores only, per Section 5.1.1). Following Eloundou et al. (2024), I report the share of workers with exposure above selected thresholds (10%, 25%, 50%, 75%) and the full density of the exposure distribution.

4.4 Digitally-reachable (readiness-adjusted) exposure

The core methodological contribution of this paper is a measure of AI exposure adjusted for digital-infrastructure readiness. This adjustment does not verify or re-measure the task content of occupations as practiced in Côte d’Ivoire — that verification is the separate, not-yet-implemented task-content check described at the end of this section (planned Specification C). What it does is scale the O*NET-calibrated exposure score by the predicted probability that a worker’s environment currently has the digital infrastructure (computer/internet access) that applying an AI tool to that task would require. For the GPT-based scores specifically, $E_i^{reachable} \leq E_i^{tech}$ by construction: this bounds the *constructed measure itself* from above, mechanically, given the score is being multiplied by a probability. **This is not a claim that Specifications A and B correctly bound *true* reachable exposure** — as detailed below and in Section 6.1, A undercounts mobile-only access while B can overcount an individual’s access via enterprise-level data, in opposite directions, so neither specification should be read as bounding the real quantity of interest, only the specific number this paper computes from it.

A circularity risk that readers should weigh explicitly. Because the exposure score E_i is already a deterministic function of ISCO code, and the digital-access probability below is predicted substantially from ISCO code, formality, and education, multiplying the two is not fully independent evidence of a “true” adjustment — some of the resulting gap between technical and digitally-reachable exposure is definitionally built into the specification rather than discovered by it. This paper reports the adjustment on that basis (a transparent, descriptive re-weighting) rather than presenting it as a causal or structurally identified estimate of task-content divergence, and treats it as complementary to, not a substitute for, the task-content verification (Specification C) described below.

Notation. For each individual i , define the technical exposure $E_i^{tech} = E_i$ (the O*NET-based score from Section 4.3), a digital-readiness probability R_i , and the digitally-reachable exposure:

$$E_i^{reachable} = E_i^{tech} \times R_i$$

where $R_i \in [0, 1]$. Two specifications for R_i are proposed.

A scale caveat, verified directly against the exposure table and not yet resolved. The multiplicative form $E_i^{tech} \times R_i$ presumes E_i^{tech} is a share or probability on $[0, 1]$, which holds for the GPT/Eloundou scores (alpha/beta/gamma $\in [0, 1]$ by construction) but not for AIOE: the ISCO-08-level AIOE scores in `isco08_ai_exposure_scores.csv` range from -2.31 to $+1.47$, with 219 of 422 codes negative (AIOE is a standardized index, not a task-share). Multiplying a negative AIOE score by $R_i < 1$ moves it *toward* zero — i.e., numerically *increases* it — which inverts the direction the readiness adjustment is meant to move the constructed measure, and can make the

digital-reachability gap (defined below) negative for AIOE, defeating its purpose as a downward-adjusted version of the technical measure. **As of this draft, the digitally-reachable exposure measure and the results in Section 5.2 should be understood as applying cleanly only to the GPT-based scores;** applying the same multiplicative form to AIOE without first placing it on a comparable non-negative scale is a known, unresolved issue, flagged here and in Section 5.4 rather than silently worked around.

Specification A — Digital access proxy:

$$R_i = \hat{P}(\text{digital access}_i \mid X_i)$$

where digital access is a binary variable indicating whether the individual uses a computer or the internet for work purposes, and X_i is a vector of controls including ISCO code, formality status, sector, region, and education. The predicted probability is estimated via a logit model on the ENE-M sample. Because X_i includes ISCO code and formality — the same variables driving E_i^{tech} and much of the paper’s discussion of informality — this specification’s relationship to formality status is presented in results (Section 5.2) as descriptive, not as evidence of a causal formality effect. **A further caveat:** “digital access” here is operationalized from ENE-M’s computer/internet-at-work variables; it does not capture smartphone-only access, which is far more widespread in Côte d’Ivoire and through which informal workers may already reach consumer AI products (as documented for the general population in Section 5.3.4’s OpenAI Signals data) without a workplace computer. Specification A therefore undercounts mobile-inclusive digital reach; it is not a full readiness measure.

The logit model allows the adjustment to vary smoothly across the formal-informal spectrum and across occupations, rather than applying a binary formal/informal indicator. Because X_i includes ISCO code, the estimation sample is implicitly restricted to the 86.9 percent of actifs occupés with a coded `citp_final` (Section 4.1); this restriction, and its potential non-randomness, is not yet characterized (Section 5.4). **A further estimation risk, not yet checked:** with several hundred possible 4-digit ISCO codes as a regressor alongside formality, sector, region, and education, many ISCO×formality cells will be sparsely populated or entirely empty (e.g., an occupation observed only among formal workers in the ENE-M sample), risking quasi-complete separation in the logit — coefficients and predicted probabilities for those cells would then be driven by estimation artifacts rather than genuine signal. This should be diagnosed (cell counts by ISCO×formality, separation checks) before Specification A’s results are reported, and is added to Section 5.4’s open items.

Specification B — Sector-formality interaction:

$$R_i = \begin{cases} 1 & \text{if formal} \\ \alpha_s & \text{if informal, sector } s \end{cases}$$

where α_s is a sector-specific scaling factor calibrated from DATAFID enterprise-level ICT adoption rates, using DATAFID’s `SECTEUR` variable — which, as noted in Section 4.1, distinguishes only three broad categories (primary/secondary/tertiary), not ISIC-section-level detail, a coarser grouping than originally planned. **A further caveat:** DATAFID’s *Entreprise* module measures *enterprise*-level equipment and connectivity (e.g., whether the business owns computers, has internet access), which is not the same as an individual informal *worker’s* access within that enterprise — the same equipment-vs-individual-access gap noted for Specification A applies here too, but in the opposite direction (an enterprise can have internet access that a given employee does not personally use,

so B can overcount where A undercounts). The $R_i = 1$ assumption for formal workers is also a simplification flagged rather than resolved: it assumes formal-sector work environments fully match the O*NET benchmark, which Specification A’s own estimates (Section 5.2.2) may show is not exactly true even for formal workers. (EHCVM remains a fallback per Section 4.1.) This specification is cruder but more transparent and does not require estimation.

Comparison and robustness. Both specifications are reported, and their implications compared. The aggregate digitally-reachable exposure is:

$$\bar{E}^{reachable} = \sum_{i=1}^N w_i \cdot E_i^{reachable}$$

The difference $\Delta = \bar{E}^{tech} - \bar{E}^{reachable}$ is termed the **digital-reachability gap** — the extent to which technical exposure exceeds the digitally-reachable exposure measure once digital-infrastructure readiness is accounted for, for the GPT-based component of the measure specifically. This is deliberately not called an “informality exposure gap”: as noted above, part of this gap is definitionally embedded in how R_i is constructed, and the term “informality” implies a claim about task content that Specifications A and B do not establish — that claim is reserved for Specification C below.

Specification C — Task-content verification (co-equal priority with A and B, not a lower-priority extension). Specifications A and B both treat the readiness adjustment as a multiplicative discount applied on top of the US-calibrated O*NET exposure score: they scale down *how much* of the assumed task bundle is currently reachable, but they do not verify *whether* the assumed task bundle itself is the right one for a given occupation as practiced in Côte d’Ivoire — which is the paper’s actual claim about task-content divergence in the informal sector (Sections 1, 3.2). The ENE-M questionnaire offers a way to check this directly: variable EP1b captures each respondent’s principal tasks and functions in free verbatim text, independently of the coded ISCO occupation (EP1b1). Specification C compares this reported task content against the O*NET task statements associated with the assigned SOC/ISCO code, for a sample of occupations where formal/informal task divergence is expected to be largest (e.g., ISCO groups 5 and 7, per Section 5.2.1). Where systematic divergence is found, this analysis will inform an adapted or reconstructed exposure score for the affected occupations, rather than relying solely on the uniform multiplicative adjustment of A/B. This task-content comparison must be conducted entirely on aggregated outputs (keyword/topic frequencies, not verbatim excerpts), consistent with the statistical confidentiality of individual ENE-M responses. **As of this draft, Specification C is not yet implemented** — no script exists for it — which means the paper’s actual task-content-divergence claim is currently supported only indirectly, by A/B’s readiness adjustment and by the qualitative discussion in Section 2.1, not by a direct test. This is stated plainly here and in Section 6.4 rather than left implicit.

4.5 Exposure-usage gap analysis (planned Kitagawa-Blinder-Oaxaca decomposition, and completed cross-provider check)

As of this draft, only the cross-provider check described at the end of this section (item 5 below) has been executed; items 1–4 describe a planned decomposition, not yet run. The final analytical component is intended to cross-reference the ENE-M-based exposure profile with observed usage data from two independent AI providers: the Anthropic Economic Index (Claude) and OpenAI Signals (ChatGPT).

Planned procedure (not yet executed):

1. From the Anthropic Economic Index data on Hugging Face, extract the distribution of Claude usage by O*NET occupation, globally and for available country-level breakdowns.
2. Construct a “usage profile” for Côte d’Ivoire: the predicted distribution of AI usage across occupations, given the country’s occupational composition, under the assumption that usage rates by occupation are equal to the global average. **This is a third instance of the transportability assumption this paper otherwise critiques** — alongside O*NET task content (Section 3.2) and U.S. employment weighting in the crosswalk (Section 4.2, 5.4) — since it borrows a cross-country-average parameter (global occupation-level usage rates) and assumes it applies to Côte d’Ivoire specifically. When this step is actually run, the same caveat given for U.S. employment weighting in Section 5.4 should be carried through here: the global-average usage profile is retained for lack of a better alternative, not because the transportability concern is resolved.
3. Compare this predicted usage profile to (a) the actual observed usage from the Anthropic data and (b) the theoretical exposure profile from the indices.
4. Apply a Kitagawa-Blinder-Oaxaca-style decomposition to separate a “composition” effect (differences in occupational structure) from a “coefficient” effect (differences in within-occupation usage rates). **This decomposition is not yet identified as a specific, estimable equation** — as Section 5.4 notes, Côte d’Ivoire’s occupation-level AEI usage is 73.6 percent unclassified (Section 5.3.1), which may leave the within-occupation “coefficient” component too imprecisely estimated to support a country-specific decomposition; a reference group, common occupational support, and treatment of AEI’s privacy suppression would all need to be specified before this step can run. Language barriers are not listed as a separable component of this decomposition: Section 5.3.3’s own test of the francophone-usage hypothesis does not reach statistical significance, so attributing part of the gap to language would not be an estimated effect but an assumption.

5. Cross-provider check (executed; results in Section 5.3.4). Compare country-level OpenAI Signals metrics — per-capita usage rank, work-related message share, and topic composition for Côte d’Ivoire — against the Anthropic Economic Index’s Claude-based picture (Section 5.3.1) and against regional peers (Ghana, Nigeria, Senegal). This is not a re-run of the items 1–4 decomposition on a second dataset — OpenAI Signals does not support occupation-level analysis outside the United States (Section 4.1) — but a validity check on whether the qualitative usage pattern found on Claude (Section 5.3.1) is a feature of AI usage in Côte d’Ivoire generally or an artifact of Claude’s specific user base; per Section 5.3.4, the two providers’ usage compositions differ, and the comparison itself is limited by the two providers’ differing populations, time windows, products, and topic taxonomies, so even this narrower check should be read as suggestive rather than as a harmonized cross-provider measurement.

Once run, the items 1–4 decomposition would be necessarily indicative rather than causal: the goal is to document the magnitude and structure of the gap, not to identify causal mechanisms.

5. Results

5.1 Standard AI exposure of the Ivorian labor force

[This section will present:]

5.1.1 Aggregate exposure. National weighted-average exposure scores under each of the two applied indices (GPT/Eloundou, AIOE) — see Section 4.1 on Pizzinelli et al.’s complementarity-

adjusted framework not yet being integrated. **The GPT and AIOE results are not directly comparable and are reported separately, not pooled or presented on a shared scale** — GPT is a $[0, 1]$ task share, AIOE a standardized index (see Section 4.4 for the full scale caveat); the 10/25/50/75 percent thresholds of Section 4.3 apply to GPT only, with an AIOE-appropriate distributional summary reported separately for AIOE. Comparison with the six-country benchmarks from Pizzinelli et al. (2023) and the 25-country results from Demombynes et al. (2025) uses those papers’ own published aggregate figures, not a re-estimation of their index on Ivorian data.

Working hypothesis, not yet tested: Côte d’Ivoire’s aggregate exposure is lower than all six countries in Pizzinelli et al., and comparable to the low-income country average in Demombynes et al., driven primarily by the occupational composition effect — the concentration of employment in ISCO groups 5, 6, and 9, which have lower exposure scores than groups 1–3. This is a prior motivated by the occupational-composition argument of Section 2.1, not a result.

5.1.2 Distribution of exposure (GPT-based scores only, per Section 5.1.1’s non-comparability note). Histogram and cumulative distribution of individual-level exposure scores. Shares of workers above the 10%, 25%, 50%, and 75% exposure thresholds (Section 4.3); an AIOE-appropriate distributional summary (standard deviation, IQR, or range — not threshold shares) will be reported separately.

Working hypothesis, not yet tested: the distribution is strongly right-skewed, with a long left tail of near-zero exposure (agricultural workers, elementary occupations) and a thin right tail of high exposure (professionals, managers, technicians concentrated in Abidjan’s formal sector).

5.1.3 Exposure by subgroup. Disaggregation by:

- **Region:** Abidjan vs. other urban vs. rural. Working hypothesis: Abidjan highest, rural lowest.
- **Sex:** Following the pattern in Pizzinelli et al. and Cazzaniga et al., women may have *higher* exposure in the formal sector (concentration in clerical and service occupations), but lower exposure overall due to higher informal employment rates.
- **Education:** A positive gradient is a plausible prior, not yet tested.
- **Age:** Younger workers in urban formal sector may have higher exposure; the pattern may be weaker or reversed in rural/informal settings.
- **Formality status:** Because technical exposure E_i^{tech} depends only on ISCO code (Section 4.3), a formal/informal gap in this section can only arise from *compositional sorting* — formal and informal workers being concentrated in different occupations — not from any direct effect of formality on the score. Working hypothesis, not yet tested: formal workers are concentrated in higher-exposure occupations. This is conceptually distinct from Section 5.2’s digital-reachability gap, which *does* depend mechanically on formality status through R_i .

Table 1: Average AI exposure by ISCO-08 major group, Côte d’Ivoire vs. benchmark countries (GPT and AIOE reported separately, per Section 5.1.1) **Figure 1:** Density plot of individual-level AI exposure (GPT-based), Côte d’Ivoire vs. South Africa (the closest emerging market benchmark) **Figure 2:** Map of average AI exposure by region (33 regions)

5.2 The digital-reachability adjustment

5.2.1 The formal-informal digital-access gap within occupations. Before applying the adjustment, I document the raw difference: for each ISCO code with sufficient observations in both the formal and informal sectors, I compare the distribution of ICT access variables. This establishes

the empirical basis for the adjustment — i.e., that formal and informal workers in the same ISCO code do, in fact, report different workplace digital-access rates. As discussed in Section 4.4, this documents a readiness gap, not on its own a task-content gap.

Working hypothesis, not yet tested: for ISCO groups 5 (services/sales) and 7 (craft workers), which are the groups with the most variation in formality status, the ICT access gap between formal and informal workers is large (e.g., [XX]% of formal ISCO 5221 workers use a computer at work vs. [XX]% of informal ISCO 5221 workers).

5.2.2 Logit model for digital access (Specification A). Estimation results for the logit model predicting workplace digital access. Key coefficients: formality status, ISCO group, sector, region, education. Marginal effects at representative values. Given the circularity caveat of Section 4.4, these coefficients are reported and interpreted descriptively — as a profile of who currently has workplace digital access — rather than as a causal formality effect.

Working hypothesis, not yet tested: formality status is a strong predictor of workplace digital access, even after controlling for occupation and education, with the predicted probability of digital access ($\hat{R}_i$) near zero for most informal workers and substantively positive only for formal workers in ISCO groups 1–4 in Abidjan.

5.2.3 Digitally-reachable exposure results. Comparison of technical and digitally-reachable aggregate exposure:

- National aggregate: $\bar{E}^{tech}$ vs. $\bar{E}^{reachable}$ under Specifications A and B
- Digital-reachability gap: $\Delta = \bar{E}^{tech} - \bar{E}^{reachable}$, expressed in absolute terms and as a percentage
- Subgroup decomposition: which groups see the largest adjustment? (Working hypothesis: rural informal workers, ISCO groups 5–9)

A construction note for Table 2’s formal/informal comparison specifically. Under Specification B, $R_i = 1$ for every formal worker and $R_i = \alpha_s < 1$ for every informal worker in sector s — so *that* formal workers show a larger $\bar{E}^{reachable}$ than informal workers in the same occupation is guaranteed by the specification, not a result Table 2 discovers. The same holds substantially for Specification A, since formality is one of its strongest predictors by design (Section 5.2.2). What Table 2 *can* inform, and what readers should look at instead of the formal/informal split itself, is the **cross-occupation and cross-region variation within each formality status** — e.g., whether the digital-reachability gap among *informal* workers varies by ISCO group or sector (via α_s) rather than being uniform, since that pattern is not mechanically fixed by the specification.

A scope note on what is and is not empirical here. For the GPT-based scores, $E_i^{tech} \in [0, 1]$ and $R_i \in [0, 1]$, so $E_i^{reachable} = E_i^{tech} \times R_i \leq E_i^{tech}$ by construction — the *constructed measure* declining is a mechanical consequence of the specification, not an empirical finding about true reachable exposure. What *is* empirical, and unknown until Sections 5.1–5.2 are populated, is the *magnitude* of the reduction and its *heterogeneity* across occupations, formality status, and region — whether the gap is a few percentage points (implying the readiness adjustment barely matters) or most of aggregate exposure (implying it matters a great deal). *Working hypothesis, not yet tested:* the reduction is [XX]–[XX] percent, negligible for Abidjan’s formal sector and large for rural informal employment; the interesting empirical content is in this pattern, not in the sign of the overall effect.

Table 2: Technical vs. digitally-reachable AI exposure by ISCO major group and formality status
Figure 3: National exposure distribution, technical vs. digitally-reachable (overlaid densities)

Figure 4: Regional map of the digital-reachability gap (Δ by region)

5.3 The exposure-usage gap

This section draws on two independent AI-provider usage datasets. Sections 5.3.1–5.3.3 use the Anthropic Economic Index (Claude), which offers occupation-level granularity but a comparatively small, professionally-skewed user base; Section 5.3.4 cross-checks the resulting picture against OpenAI Signals (ChatGPT), whose far larger and more consumer-oriented Ivorian user base cannot be broken down by occupation outside the United States but provides a country-level check on whether patterns found on Claude generalize.

5.3.1 Observed AI usage in Côte d’Ivoire. Using the Anthropic Economic Index (AEI) country-level release (`aei_enriched_claude_ai_country_2025-08.csv`, week of 2025-08-04), Côte d’Ivoire clears the Index’s minimum-observations privacy threshold (200 conversations), with 939 conversations recorded in the observation window — enough for country-level but not for reliable occupation-level breakdowns (see below). Côte d’Ivoire’s usage-per-capita index — usage relative to what its share of world population would predict, with 1.0 denoting proportional usage — is 0.199, placing it in the lowest usage quartile (“Tier 1”) alongside Nigeria (0.202), Senegal (0.295), and Ghana (0.348), and far below the United Kingdom (2.665) and United States (3.624). The automation/augmentation split for Côte d’Ivoire is 59.6 percent automation-oriented interactions (directive, feedback-loop patterns) versus 40.4 percent augmentation-oriented (validation, task-iteration, learning patterns) — a more automation-leaning split than is typically reported for high-income countries, consistent with the pattern documented by Fan and Nguyen (2026). I have not yet reproduced Fan and Nguyen’s AI Concentration Index (ACI) methodology directly; the SOC-occupation breakdown reported next provides a simpler, complementary signal of concentration.

At the occupation level, AEI data for Côte d’Ivoire is sparse: only two SOC major groups meet the minimum-observations threshold — Computer and Mathematical (24.6 percent of total usage) and Educational Instruction and Library (1.8 percent) — with the remaining 73.6 percent of usage falling into “not_classified” (suppressed for insufficient granularity); the three figures sum to 100 percent of total usage. **This 73.6 percent is precisely unknown, not zero or evenly distributed** — it could in principle be concentrated in the same IT-adjacent occupations as the classified share, or spread thinly across dozens of low-volume occupations, or anywhere in between, and the AEI’s privacy suppression does not let this be distinguished with the current sample. Consequently, the *minimum* possible concentration in Computer/Mathematical plus Educational Instruction is 26.4 percent (if none of the unclassified share falls there), and the *maximum* is 100 percent (if all of it does); the true figure is unknown within that range. The O*NET tasks that do surface among the classified 26.4 percent are dominated by software development activities (writing and modifying code, web development, troubleshooting) plus one non-technical task (assisting students with coursework). This is directionally consistent with — but, given the width of the interval above, cannot on its own establish — the “narrow professional enclave” pattern described by Fan and Nguyen (2026); it is reported here as suggestive of that pattern within the classified share, not as a measurement of overall concentration.

5.3.2 Predicted vs. actual usage. The Kitagawa-Blinder-Oaxaca decomposition proposed in Section 4.5 — separating the occupational-composition effect from the within-occupation usage gap — has not yet been implemented. The crosswalk employment-weighting refinement it was previously waiting on (Section 4.2) is now complete; the decomposition itself, merging the full occupation-level AEI usage profile against the ENE-M occupational distribution, remains the next

step. The sparse occupation-level AEI coverage documented in 5.3.1 (73.6 percent not_classified) suggests the within-occupation component may be difficult to estimate precisely for Côte d’Ivoire specifically, and the decomposition may need to rely more heavily on the cross-country regularities documented in 5.3.3 than on Côte d’Ivoire-specific occupation cells.

5.3.3 The language dimension. Extracting all 48 Sub-Saharan and North African countries classified as Francophone, Anglophone, Lusophone, or Arabophone by dominant official language, 43 are present in the AEI data with non-zero usage. Descriptively, mean usage-per-capita index is 0.450 for Arabophone countries, 0.260 for Anglophone, 0.139 for Francophone, and 0.115 for Lusophone countries. Regressing the usage-per-capita index on log GDP per working-age capita and a Francophone indicator, restricted to the 33 Francophone and Anglophone countries, yields a positive and highly significant GDP coefficient (0.182, $p < 0.001$) and a negative Francophone coefficient (-0.080 , $p = 0.23$, $R^2 = 0.42$). **At $p = 0.23$, this result does not reach conventional statistical significance, and it is reported as such: controlling only for GDP, the data here do not provide statistically significant evidence of a Francophone-specific usage disadvantage.** The coefficient’s sign and rough magnitude are directionally compatible with the language-barrier hypothesis of Fan and Nguyen (2026), but with only two controls (GDP and a single language dummy) and $n = 33$, the specification cannot separate a language effect from correlated omitted factors (connectivity, education, colonial-era institutions), so this remains, at most, a hypothesis not rejected by the data rather than a finding the paper argues for. Notably, Côte d’Ivoire’s own usage-per-capita index (0.199) sits above the Francophone-country median (0.117), indicating it is not a particularly low-usage outlier within its linguistic peer group.

5.3.4 Cross-provider check: OpenAI Signals. As motivated in Sections 3.3 and 4.5, Section 5.3.1’s classified-usage pattern may reflect Claude’s specific, professionally-skewed Ivorian user base rather than a general property of AI usage in Côte d’Ivoire; OpenAI Signals provides a country-level (not occupation-level) point of comparison from a much larger and more consumer-oriented user base. Three metrics are compared, computed by `scripts/16_analyser_openai_signals_civ.R` from the public OpenAI Signals release (Section 4.1): (i) **per-capita usage rank** — Côte d’Ivoire ranks 118th–125th of 147 countries with complete estimates across the six quarters from Q1 2025 to Q2 2026 (125th in the latest quarter), a similarly low position to its AEI Tier-1 (lowest-quartile) usage-per-capita ranking, placing it behind Ghana (112th) and Nigeria (116th) in the most recent quarter; (ii) **work-relatedness** — 28.7 percent of Côte d’Ivoire’s ChatGPT messages are classified work-related in the latest month, close to the 30.4 percent global average, which is itself a useful data point since it means ChatGPT usage in Côte d’Ivoire is not unusually leisure-skewed relative to the rest of the world; (iii) **topic composition** — the largest categories are Writing (34.6 percent) and Practical Guidance (29.9 percent), with Technical Help at only 2.6 percent; this is a topic-based breakdown, not directly comparable to the occupation-based classified share of Claude usage documented in 5.3.1 (see the discussion below), though the two paint qualitatively different pictures on their face.

This third metric is the one that must be interpreted carefully. OpenAI Signals does not publish occupation-level (O*NET/SOC) breakdowns for Côte d’Ivoire, so the 2.6 percent Technical Help share cannot be mapped onto ISCO or SOC occupations, and it is not itself a measure of occupational concentration — a narrow professional enclave could in principle still write disproportionately using ChatGPT for non-technical purposes (drafting reports, correspondence) without this showing up as “Technical Help.” What Section 5.3.4 therefore supports is a narrower and safer claim than “the Claude enclave pattern does not hold on ChatGPT”: **ChatGPT’s topic breakdown and Claude’s classified occupation mix are different, non-directly-comparable descriptors of**

usage — one categorizes by conversational topic, the other by O*NET/SOC occupation — and the two providers’ usage compositions should not be pooled or treated as interchangeable measures of “AI usage in Côte d’Ivoire” on the strength of this comparison alone. Whether the qualitative difference between an IT-skewed occupation mix (Claude) and a writing/guidance-skewed topic mix (ChatGPT) reflects two different slices of the same underlying (occupationally concentrated) population, or two populations that differ in occupational composition as well as topic, cannot be determined from the data available here.

Table 3: AI usage metrics, Côte d’Ivoire vs. selected African and emerging-market benchmarks (Claude/AEI and ChatGPT/OpenAI Signals, side by side) **Figure 5:** Scatter plot of AI exposure (x-axis) vs. observed AI usage (y-axis) by occupation, Côte d’Ivoire vs. United States (Claude/AEI only, per the occupation-level data limitation of OpenAI Signals noted above)

5.4 Robustness and measurement caveats

The choices made throughout Sections 4–5 are not innocuous defaults; several are addressed here together, rather than scattered across appendices, because they bear directly on how the results in 5.1–5.3 should be read.

Resolved and quantified: - *Crosswalk aggregation weighting.* OEWS employment weighting vs. unweighted mean shifts individual ISCO-08 exposure scores non-trivially (mean absolute difference 0.017 GPT-alpha / 0.073 AIOE, correlation 0.956/0.990) — Appendix D. - *GPT/AIOE match-rate gap: diagnosis and coverage bug.* What is resolved: the *cause* of the gap is traced to a SOC-2010/SOC-2018 vintage mismatch between the two indices and the crosswalk, not random non-coverage, and the exposure table now uses the union of GPT and AIOE matches (423 of 436 ISCO-08 codes) rather than the GPT-only support (392 codes) that silently dropped AIOE-only codes — Appendix A. **What is not resolved:** the underlying vintage mismatch itself (no SOC-2010/2018 bridge crosswalk has been sourced), listed below. - *DATAFID sectoral granularity.* SECTEUR distinguishes only primary/secondary/tertiary, not ISIC sections as originally planned for Specification B — Section 4.1, 4.4.

Flagged but not yet resolved — planned for the next revision: - *SOC 2010 ↔ SOC 2018 harmonization* — affects both the crosswalk and the OEWS weights. Two distinct instances of the same underlying problem: (i) GPT scores vs. the SOC-2010 crosswalk (Appendix A), and (ii) the May 2020 OEWS employment file itself, which turns out to use post-2018 SOC codes rather than SOC-2010 as this paper initially assumed (Section 4.1) — weight coverage is 95.8 percent for GPT-matched (post-2018-consistent) pairs but only 83.6 percent for AIOE-matched (SOC-2010) pairs, the reverse of what a SOC-2010-vintage OEWS file would produce. Both require sourcing correctly-vintaged BLS files (subject to the same access constraint noted in Section 4.1). - *AIOE scale incompatibility with the multiplicative readiness adjustment.* AIOE scores range from -2.31 to $+1.47$ (219 of 422 ISCO-08 codes negative) — a standardized index, not a $[0, 1]$ task share — so $E_i^{tech} \times R_i$ does not behave as intended for AIOE-based exposure (Section 4.4): multiplying a negative score by $R_i < 1$ moves it toward zero, i.e., increases it. Sections 5.1–5.2’s digitally-reachable exposure results should be read as applying cleanly to the GPT-based scores only until this is resolved, either by restricting the adjustment to GPT or by placing AIOE on a comparable non-negative scale first. - *Panel pooling.* Person-quarters vs. distinct individuals, survey design (strata/PSU), and within-person correlation across the seven pooled quarters are not yet accounted for in point estimates or standard errors (Section 4.1). This is a substantive, not merely technical, gap: if respondents retained across more quarters differ systematically in

formality or region from those who attrite after one quarter (plausible if formal-sector jobs are more stable), the pooled sample’s occupational and formality composition — and therefore the aggregate exposure estimates in Sections 5.1–5.2 — could be biased relative to the true cross-sectional population, not just imprecisely estimated. - *The ENE-M estimand is not yet pinned down.* Section 4.3 defines aggregates using survey weights, but Section 4.1 notes the panel-pooling and weighting issue above is unresolved; whether the reported aggregates will represent a single-quarter estimate, an average across the seven pooled quarterly populations, or a deduplicated-individual estimate is not yet specified, and the choice affects both point estimates and their interpretation as “nationally representative.” - *Uncertainty quantification generally.* Results in Sections 5.1–5.3 are presented as point estimates. None of the following sources of uncertainty is yet propagated into confidence intervals: ENE-M sampling error, Specification A/B parameter uncertainty, DATAFID sampling error, or crosswalk ambiguity. Table and figure captions should not use the word “significant” until this is addressed for the relevant estimate. - *Section 2’s institutional statistics need inline citations.* The GDP-per-capita, growth-rate, informal-employment-rate (89–93 percent), and ENSESI 2016 SU4 figures in Section 2.1 are stated without inline source citations; they should be traced to the specific ENE/ENSESI/national accounts releases they come from before this draft circulates beyond internal use. - *Unmatched-code and non-coded-occupation selection.* The 13 ISCO-08 codes with no exposure score under either index are currently dropped rather than imputed or bounded, and — separately — the 13.1 percent of actifs occupés without a coded `citp_final` (Section 4.1) are not yet characterized by formality, region, sector, or education. Both exclusions are plausibly non-random (concentrated in agricultural/artisanal and technology-renumbered occupations for the former; unclear but unverified for the latter) rather than a source of bias in no particular direction as an earlier version of this section claimed; their ENE-M employment shares and a sensitivity range are not yet computed. - *The crosswalk’s promised diagnostics are incomplete.* Section 4.2 states that the coefficient of variation of exposure scores within each ISCO code will be reported as a crosswalk-quality diagnostic; this has not yet been computed or added to Appendix A. For AIOE specifically, a coefficient of variation is not a meaningful summary given means near zero or negative (see the AIOE-scale item above) — standard deviation, interquartile range, or range should be used for AIOE instead of a mechanically-applied CV. - *US employment weighting reintroduces a transportability assumption the paper otherwise critiques.* Weighting SOC-to-ISCO aggregation by U.S. employment shares (Section 4.2) assumes the sub-occupational composition within an ISCO-08 code in Côte d’Ivoire mirrors that of the United States — structurally the same transportability assumption Section 3.2 argues against for task content generally. The weighted/unweighted sensitivity in Appendix D is informative but does not resolve this; U.S. weighting is retained as the primary specification pending a better alternative (e.g., ENE-M-based weights, where occupational detail permits), not because the concern is resolved. - *Specification A/B circularity and interpretation.* Addressed by renaming and reframing (Section 4.4) rather than by resolving the underlying identification problem; Specification C (task-content verification, Section 4.4) is the paper’s actual planned resolution and remains unimplemented. - *Specification A’s logit may face sparse-cell/separation problems.* With several hundred possible ISCO codes as a regressor alongside formality, sector, region, and education, some ISCO×formality cells are likely to be sparsely populated or empty, risking quasi-complete separation; cell counts and a separation diagnostic have not yet been run (Section 4.4). - *Table 2’s formal/informal comparison is largely mechanical.* Under Specification B, and substantially under A, formal workers showing a larger digitally-reachable exposure than informal workers in the same occupation follows from the specifications’ construction, not from data; the informative content of Table 2 is in cross-occupation/cross-region variation *within* formality status, not the formal/informal split itself (Section 5.2.3). - *Pizzinelli et al. (2023) and the Gmyrek et al. (2025) ILO index.* Neither is integrated as an applied measure in this draft (Section 4.1, 3.1); the technical

(unadjusted) exposure results planned for Section 5.1 rest on GPT and AIOE, reported separately (Section 5.1.1); the digitally-reachable exposure results planned for Section 5.2 rest on GPT only, per the AIOE-scale item above. - *Oaxaca-Blinder decomposition (Section 4.5)*. Not yet run and not yet fully identified as an estimable equation — the reference group, common occupational support, and treatment of AEI’s privacy suppression still need to be specified (Section 4.5, 5.3.2); the crosswalk weighting it was previously blocked on is now complete, but is not the only remaining obstacle. Its Step 2 also imports a third instance of the transportability assumption critiqued elsewhere in this paper (global-average usage rates by occupation applied to Côte d’Ivoire) — see Section 4.5. - *Cross-provider (AEI/Signals) comparability*. The two datasets differ in population, time window, product/plan coverage, privacy treatment, and topic taxonomy (Section 5.3.4); the country-level comparison reported there is not a harmonized measurement and should not be extended beyond the narrow claim it supports.

Given the number of items above still open — several substantive, not merely presentational — the point estimates that *will* populate Sections 5.1–5.2’s [XX] placeholders should be read, on first release, as a first empirical pass rather than as final headline numbers. That the GPT-based constructed measure declines under the readiness adjustment is mechanical (Section 5.2.3), not something contingent on the items above; what those items do bear on is the *magnitude and pattern* of that decline, and whether it says anything reliable about true reachable exposure rather than only about the constructed measure — readers should not treat this draft’s eventual point estimates as precise, or as evidence about the real world beyond the constructed measure itself, until the panel-pooling, AIOE-scale, and SOC-vintage items above are resolved.

6. Discussion

6.1 Implications for measurement

The results challenge the straightforward application of AI exposure indices to countries with dual labor markets. The digital-reachability adjustment proposed in this paper (Section 4.4) is simple and replicable — it requires only a labor force survey with ISCO codes, formality indicators, and ICT access variables, all of which are standard in modern labor force surveys in sub-Saharan Africa (including the harmonized surveys of the WAEMU zone) — though, as Section 4.4 and 5.4 make explicit, it measures infrastructure readiness rather than task content, and should not be conflated with a direct informality adjustment until the task-content check (Specification C) is complete. The key message for the measurement community — including the ILO, which maintains the ISCO classification, and the World Bank, which supports labor force survey harmonization — is twofold: for the GPT-based (task-share) component of exposure, applying a readiness adjustment mechanically produces a smaller *constructed measure* (Section 5.2.3), which is suggestive that unadjusted occupation-level exposure scores overstate what is currently reachable in economies with limited digital infrastructure — but only suggestive, since this paper’s own readiness adjustments (Section 4.4) are not uniformly reliable proxies for true reachable exposure: Specification A undercounts mobile-only access while Specification B’s enterprise-level ICT data can overstate an individual worker’s actual access, in opposite directions. Neither should be read as a precise correction, and neither substitutes for verifying whether the underlying task-content assumption (Section 3.2) holds in the first place.

This finding speaks directly to the ongoing debate in the AI-and-labor literature about the gap between theoretical exposure and realized impact. Acemoglu (2024) has argued that the realized macroeconomic impact of AI will be substantially smaller than the exposure scores suggest, even

in advanced economies, due to the cost and difficulty of implementing AI in real work contexts, and Hampole, Papanikolaou, Schmidt, and Seegmiller (2025) provide direct labor market evidence consistent with this pattern: task-level AI exposure demonstrably reduces labor demand for exposed tasks, yet the aggregate employment consequences they find are surprisingly modest. The present paper extends this argument to developing economies with a working hypothesis, not yet tested: that the gap between theoretical exposure and effective exposure is larger there, amplified by a structural factor — labor market informality — that the existing literature has not addressed.

6.2 Implications for policy

For Ivorian policymakers, the results carry a dual message, which should be read with the scope limits of Section 5.4 in mind: this paper measures current digital reachability and observed usage, not future labor market outcomes, employment, wages, or productivity. On one hand, the low digitally-reachable exposure of the bulk of the labor force, if confirmed once Sections 5.1–5.2 are populated, would suggest that *currently reachable* AI-driven disruption is modest — most Ivorian workers currently lack the workplace digital infrastructure the exposure indices implicitly assume. This is a statement about present-day infrastructure constraints, not a prediction: falling device costs, smartphone-based AI access, and platform-mediated task delivery could all expand effective reach without any change in occupational structure, so low current readiness should not be read as evidence of low future exposure. This point qualifies, rather than contradicts, the more alarming headlines from global reports that emphasize the share of employment “exposed” to AI without adjusting for present-day infrastructure.

On the other hand, the concentration of Claude usage in a narrow, IT-adjacent segment of observed activity (Section 5.3.1) — a pattern that does not straightforwardly extend to ChatGPT’s topic composition (Section 5.3.4) — raises distributional questions worth monitoring even though this paper cannot yet establish whether it reflects occupational concentration, provider-specific user-base composition, or both. If AI productivity gains, whatever their eventual scale, do concentrate disproportionately among the minority with reliable formal-sector digital access, the already wide productivity gap between formal and informal employment could widen further. This is consistent with, without being independently confirmed by, the Anthropic Economic Index’s broader finding that AI usage in developing economies is concentrated in a small professional enclave (Fan & Nguyen, 2026).

The policy implication is not that AI can be ignored in countries like Côte d’Ivoire, but that the relevant policy questions are different from those in advanced economies. Rather than asking “how do we retrain workers displaced by AI?”, the more pertinent questions are: (i) how can the productivity benefits of AI be extended beyond the formal-sector enclave? (ii) what infrastructure investments (electricity, connectivity, digital literacy) are prerequisites for broader AI diffusion? and (iii) how should educational systems prepare for a labor market where the digital-informal divide may become the dominant axis of inequality?

6.3 Linking provider usage data to national labor force surveys

More generally, this paper’s approach — juxtaposing publicly released AI-provider usage data against a national labor force survey’s occupational and formality detail, and, once Section 4.5’s planned decomposition is run, directly linking the two — is not specific to Côte d’Ivoire or to either provider used here. Section 5.3.4’s finding that Claude and ChatGPT report usage on such different descriptors for the same country suggests that this kind of comparison is more informative, and

more prone to provider-specific artifacts, than relying on a single provider’s usage release would suggest. Extending this approach to other countries would require the same ingredients used here: a labor force survey with detailed occupational coding and formality status, and, ideally, usage data from more than one AI provider.

For francophone Africa specifically, this paper highlights a research agenda that is currently underserved: understanding how AI diffuses in economies where the dominant employment form is informal, the dominant language is not English, and the dominant infrastructure constraints are not bandwidth but electricity and device access.

6.4 Limitations

Several limitations should be noted beyond the measurement-specific items already consolidated in Section 5.4. First, the exposure indices used in this paper were developed to measure the capabilities of GPT-4-class models as of 2023–2024. AI capabilities are advancing rapidly, and exposure measures will need to be updated as models become more capable, multilingual, and multimodal. Second, the digital-reachability adjustment (Section 4.4) relies on ICT access as a proxy for infrastructure readiness, not for task content — a distinction this paper now maintains explicitly (Section 4.4, 5.4) after initially conflating the two — and even as a readiness proxy it is imperfect: some informal workers may reach AI tools via smartphone without a workplace computer (Section 4.4’s Specification A caveat), and some formal workers with digital access may not use AI even if their tasks are theoretically exposed. Third, the Anthropic Economic Index data for Côte d’Ivoire is sparse at the occupation level, and the cross-provider comparison with OpenAI Signals (Section 5.3.4) is itself limited by the absence of occupation-level ChatGPT data for Côte d’Ivoire; both usage-side findings should be read as suggestive rather than definitive pending richer data. Fourth, the paper is descriptive rather than causal: it documents patterns of exposure and usage, but does not identify causal effects of AI on employment, wages, or productivity in the Ivorian context, and the policy discussion in Section 6.2 is framed accordingly, as a statement about current reachability rather than a forecast. Such causal analysis would require either experimental variation in AI access or quasi-experimental identification strategies, which are left for future work.

7. Conclusion

This paper maps AI occupational exposure onto a francophone West African economy’s labor force using ISCO-08-coded micro-data from Côte d’Ivoire’s ENE-M — to my knowledge, the first application of the GPT and AIOE exposure indices at this level of occupational detail for a Sub-Saharan Francophone country, though the broader literature on developing-country AI exposure (Section 3) is moving quickly enough that this claim should be understood as bounded by what was known to the author at the time of writing (August 2026), not as a permanent priority claim. For the GPT-based scores, that the digitally-reachable exposure measure (Section 4.4) is no larger than technical exposure follows mechanically from its construction (Section 5.2.3) — that part is not in question. What Sections 5.1–5.2’s results, once populated, will speak to is the *magnitude and distribution* of that mechanical decline across occupations, formality status, and region, and — separately, and only as reliably as Specifications A and B’s imperfect readiness proxies allow (Section 4.4) — whether this pattern is informative about *true* currently-reachable exposure rather than only about the constructed measure. The task-content check (Specification C), not yet implemented, remains the more direct test of the paper’s underlying claim about task-content divergence in the informal sector.

The gap between theoretical exposure and observed AI usage, documented using two independent AI providers (the Anthropic Economic Index and OpenAI Signals), further illustrates the distance between what AI *could* do and what AI *is being used for* in the Ivorian economy — and, comparing the two providers, that they report this distance using different, non-directly-comparable descriptors (occupation for Claude, topic for ChatGPT), so which provider’s usage data is used can change the qualitative picture that emerges. Occupational composition and digital-infrastructure readiness plausibly dampen the effective diffusion of current AI systems in the Ivorian labor market; a francophone-language factor remains, at this stage, an untested hypothesis rather than a documented mechanism — the data in Section 5.3.3 ($p=0.23$, $n=33$) are too imprecise to distinguish a null effect from an economically meaningful one, and it is listed here as a candidate for future work, not as one of this paper’s findings. This paper does not yet decompose the relative contributions of the mechanisms it does document (Section 5.3.2).

These findings, once complete, would contribute to a growing body of evidence that the labor market impact of AI is likely to be uneven across countries, shaped not only by occupational structure but also by digital-infrastructure readiness — a dimension the current generation of exposure indices does not capture. The readiness adjustment proposed here is one step toward more context-appropriate measurement; the task-content verification planned as Specification C (Section 4.4) is the more direct test of the paper’s underlying claim about task-content divergence in the informal sector, and is a priority for the next revision alongside the other open items in Section 5.4. As AI capabilities advance and diffuse more broadly, repeated application of this framework to successive rounds of the ENE-M panel would allow monitoring of how digitally-reachable AI exposure evolves over time in a dual labor market.

Acknowledgments

This draft’s methodology was revised following an independent multi-model critical review that identified several issues corrected in this version — most substantively, the SOC-vintage mismatches documented in Appendix A and Section 4.1, the circularity risk in Specifications A/B addressed by reframing them as a digital-readiness rather than task-content measure, and the cross-provider usage claims narrowed in Section 5.3.4. This is noted here for transparency about the paper’s current state.

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Appendix A: ISCO-08 to O*NET-SOC crosswalk details

The IBS crosswalk links 839 unique O*NET-SOC codes to 436 unique ISCO-08 codes (1,131 SOC–ISCO pairs). Match rates differ by index: 80.0 percent of pairs match a GPT (Eloundou et al. 2024) score, 91.8 percent match an AIOE (Felten et al. 2021) score, and 92.1 percent match at least one of the two. At the ISCO-08 level, this yields exposure scores for 392 codes under GPT, 422 under AIOE, and **423 under the union of the two — this union is what the published exposure table (`isco08_ai_exposure_scores.csv`) now reports** (corrected 2026-08-13; an earlier version of `scripts/03_construire_crosswalk_exposition_isco.R` filtered on non-missing GPT scores only, silently dropping the roughly 30 ISCO-08 codes that have an AIOE match but no GPT match — fixed by widening the filter to GPT-or-AIOE). Only 13 of the 436 ISCO-08 codes in the crosswalk (3.0 percent) have no score under either index and are treated as missing rather than imputed. Among the SOC codes that do carry a GPT exposure score, 95.8 percent also carry a BLS OEWS

national employment weight (Section 4.1); the remaining 4.2 percent are excluded from the weighted average (effectively zero-weighted) within their ISCO-08 code, except where *no* matched SOC code in that ISCO-08 code carries a weight, in which case the unweighted mean is used for that code only.

A more consequential issue than the match rate itself is *why* GPT and AIOE match at such different rates: the two indices use different O*NET-SOC vintages, which the numeric soc10 merge in scripts/03_construire_crosswalk_exposition_isco.R does not detect. Felten et al.’s AIOE codes are SOC 2010-era (e.g., 15-1131 “Computer Programmers” is present and matches the IBS crosswalk, which is itself built on O*NET-SOC 2010 per its source do-file — the crosswalk package documents having last been run against the “20.1 O*NET release, 2015”). Eloundou et al.’s GPT scores use the post-2018 SOC revision (e.g., 15-1251 “Computer Programmers”, 15-1299.07 “Blockchain Engineers” — codes introduced by the 2018 restructuring of the Computer Occupations group and entirely absent from the SOC-2010-vintage crosswalk; verified directly against both files). Occupations renumbered or newly created between SOC 2010 and SOC 2018 — concentrated in exactly the technology-adjacent occupations most relevant to AI exposure — are therefore silently unmatched on the GPT side specifically. A numeric-code merge across nomenclature vintages is also a latent risk of *silent mismatches* (two unrelated occupations sharing a recycled code number across vintages), not only missed matches, though no instance of this has been confirmed in the current data. Reconciling this properly requires a SOC 2010 ↔ SOC 2018 bridge crosswalk (available from the BLS, subject to the same automated-access block as the other BLS files used in this paper — a public-data mirror would need to be located, as was done for the OEWS employment weights in Section 4.1) inserted between the GPT scores and the ISCO crosswalk. Until this is done, GPT-based exposure scores for technology-adjacent ISCO-08 codes should be treated as a lower-confidence, likely under-covered component of the exposure measure; AIOE, being vintage-consistent with the crosswalk, is not affected by this specific issue.

[To be completed: full concordance table by ISCO-08 code; SOC 2010/2018 harmonization once a bridge crosswalk is sourced]

Appendix B: Exposure scores by 2-digit ISCO-08

[To be completed: table of GPT and AIOE scores for each 2-digit ISCO group, with ENE-M employment shares; Pizzinelli complementarity scores to be added if/when that index is integrated]

Appendix C: Regional exposure maps

[To be completed: choropleth maps of exposure by region, unadjusted and adjusted]

Appendix D: Sensitivity analysis

Crosswalk aggregation: employment-weighted vs. unweighted mean. Comparing the OEWS employment-weighted ISCO-08 exposure scores (Section 4.2) against an unweighted mean across matched SOC codes, on the 392 ISCO-08 codes with GPT-matched support (the scope of the pre-correction backup file — see Appendix A on the subsequent widening to 423 codes): the two are highly correlated ($r = 0.956$ for the GPT alpha score, $r = 0.990$ for AIOE), but individual codes can shift substantially — mean absolute difference of 0.017 (GPT alpha, on a 0–1 scale) and 0.073 (AIOE), with a maximum absolute difference of 0.377 (GPT alpha) and

0.898 (AIOE) for the ISCO codes whose matched SOC codes have the most unequal U.S. employment shares. This confirms that the unweighted aggregation used in earlier iterations of this pipeline was not innocuous for specific occupations, motivating the switch to employment weighting as the primary specification. Results using the unweighted alternative are available in `data/processed/isco08_ai_exposure_scores_UNWEIGHTED_backup.csv` for direct comparison. This comparison should be re-run against the full 423-code union once feasible, though there is no reason to expect the qualitative conclusion (weighting matters for specific codes, is not innocuous) to change.

[To be completed: (ii) exclusion of unmatched occupations, (iii) alternative digital-reachability adjustment specifications]